CHAPTER 16

How to Design Effective Tables

A tabular presentation of data is often the heart or, better, the brain, of a scientific paper.

—Peter Morgan

WHEN TO USE TABLES

Before proceeding to the "how to" of tables, let us first examine the question of "whether to."

As a rule, do not construct a table unless repetitive data *must* be presented. There are two reasons for this general rule. First, it is simply not good science to regurgitate reams of data just because you have them in your laboratory notebooks; only samples and breakpoints need be given. Second, the cost of publishing tables can be high compared with that of text, and all of us involved with the generation and publication of scientific literature should worry about the cost.

If you made (or need to present) only a few determinations, give the data in the text. Tables 1 and 2 are useless, yet they are typical of many tables that are submitted to journals.

Table 1 is faulty because two of the columns give standard conditions, not variables and not data. If temperature is a variable in the experiments, it can have its column. If all experiments were done at the same temperature, however, this single bit of information should be noted in the materials and methods section and perhaps as a footnote to the table, but not in a column in the table. The data presented in the table can be given in the text itself in a form that is readily comprehensible to the reader, without taking up space with a table. Very simply, these results would read: "Aeration of the growth medium

Table 1. Effect of aeration on growth of *Streptomyces coelicolor*

Temp (°C)	No. of expt	Aeration of growth medium	Growth[a]
24	5	+[b]	78
24	5	−	0

[a]As determined by optical density (Klett units).
[b]Symbols: +, 500-ml Erlenmeyer flasks were aerated by having a graduate student blow into the bottles for 15 min out of each hour; −, identical test conditions, except that the aeration was provided by an elderly professor.

was essential for the growth of *Streptomyces coelicolor.* At room temperature (24°C), no growth was evident in stationary (unaerated) cultures, whereas substantial growth (OD, 78 Klett units) occurred in shaken cultures."

Table 2 has no columns of identical readings, and it looks like a good table. But is it? The independent variable column (temperature) looks reasonable enough, but the dependent variable column (growth) has a suspicious number of zeros. You should question any table with a large number of zeros (whatever the unit of measurement) or a large number of 100s when percentages are used. Table 2 is a useless table because all it tells us is that "The oak seedlings grew at temperatures between 20 and 40°C; no measurable growth occurred at temperatures below 20°C or above 40°C."

In addition to zeros and 100s, be suspicious of plus and minus signs. Table 3 is of a type that often appears in print, although it is obviously not very informative. All this table tells us is that "*S. griseus, S. coelicolor, S. everycolor,* and *S. rainbowenski* grew under aerobic conditions, whereas *S. nocolor* and *S. greenicus* required anaerobic conditions." Whenever a table, or columns within a table, can be readily put into words, do it.

Some authors believe that all numerical data must be put in a table. Table 4 is a sad example. It gets sadder when we learn (at the end of the footnote) that the results were not statistically significant anyway ($P = 0.21$). If these data were worth publishing (which seems doubtful), one sentence in the results would have done the job: "The difference between the failure rates—14 percent (5 of 35) for nocillin and 26 percent (9 of 34) for potassium penicillin V—was not significant ($P = 0.21$)."

In presenting numbers, give only significant figures. Nonsignificant figures may mislead the reader by creating a false sense of precision; they also make comparison of the data more difficult. Unessential data, such as laboratory numbers, results of simple calculations, and columns that show no significant variations, should be omitted.

Another very common but often useless table is the word list. Such a list might be suitable for a slide in a presentation, but it does not belong in a scientific paper. Table 5 is an example. This information could easily be presented

Table 2. Effect of temperature on growth of oak (*Quercus*) seedlings[a]

Temp (°C)	Growth in 48 h (mm)
−50	0
−40	0
−30	0
−20	0
−10	0
0	0
10	0
20	7
30	8
40	1
50	0
60	0
70	0
80	0
90	0
100	0

[a]Each individual seedling was maintained in an individual round pot, 10 cm in diameter and 100 cm high, in a rich growth medium containing 50% Michigan peat and 50% dried horse manure. Actually, it wasn't "50% Michigan"; the peat was 100% "Michigan," all of it coming from that state. And the manure wasn't half-dried (50%); it was all dried. And, come to think about it, I should have said "50% dried manure (horse)"; I didn't dry the horse at all.

Table 3. Oxygen requirement of various species of *Streptomyces*

Organism	Growth under aerobic conditions[a]	Growth under anaerobic conditions
Streptomyces griseus	+	−
S. coelicolor	+	−
S. nocolor	−	+
S. everycolor	+	−
S. greenicus	−	+
S. rainbowensky	+	−

[a]See Table 1 for explanation of symbols. In this experiment, the cultures were aerated by a shaking machine (New Brunswick Shaking Co., Scientific, NJ).

Table 4. Bacteriological failure rates

Nocillin	K Penicillin
5/35 (14)[a]	9/34 (26)

[a]Results expressed as number of failures/total, which is then converted to a percentage (within parentheses). $P = 0.21$.

Table 5. Adverse effects of nicklecillin in 24 adult patients

No. of patients	Side effect
14	Diarrhea
5	Eosinophilia (≥5 eos/mm^3)
2	Metallic taste[a]
1	Yeast vaginitis[b]
1	Mild rise in urea nitrogen
1	Hematuria (8–10 rbc/hpf)

[a]Both of the patients who tasted metallic worked in a zinc mine.
[b]The infecting organism was a rare strain of *Candida albicans* that causes vaginitis in yeasts but not in humans.

in the text. A good copyeditor will kill this kind of table and incorporate the data into the text. Yet, when copyeditors do so (and this leads to the next rule about tables), they often find that much or all of the information was already in the text. Thus, the rule: Present the data in the text, or in a table, or in a figure. *Never* present the same data in more than one way. Of course, selected data can be singled out for discussion in the text.

Tables 1 to 5 provide typical examples of the kinds of material that should not be tabulated. Now let us look at material that should be tabulated.

HOW TO ARRANGE TABULAR MATERIAL

Having decided to tabulate, you ask yourself the question: "How do I arrange the data?" Since a table has both left-right and up-down dimensions, you have two choices. The data can be presented either horizontally or vertically. But *can* does not mean *should;* the data should be organized so that the like elements read *down,* not across.

Examine Tables 6 and 7. They are equivalent, except that Table 6 reads across, whereas Table 7 reads down. To use an old fishing expression, Table 6 is "bass ackward." Table 7 is the preferred format because it allows the reader to grasp the information more easily, and it is more compact and thus less expensive to print. The point about ease for the reader would seem to be obvious. (Did you

ever try to add numbers that were listed horizontally rather than vertically?) The point about reduced printing costs refers to the fact that all columns must be wide or deep in the across arrangement because of the diversity of elements, whereas some columns (especially those with numbers) can be narrow without runovers in the down arrangement. Thus, Table 7 appears to be smaller than Table 6, although it contains the same information.

Words in a column are lined up on the left. Numbers are lined up on the right (or on the decimal point). Table 7, for example, illustrates this point.

Table 8 is an example of a well-constructed table (reprinted with permission from the instructions to authors of the *Journal of Bacteriology*). It reads down, not across. It has headings that are clear enough to make the meaning of the data understandable without reference to the text. It has explanatory footnotes, but they do not repeat excessive experimental detail. Note the distinction here. It is proper to provide enough information so that the meaning of the data is apparent without reference to the text, but it is improper to provide *in the table* the experimental detail that would be required to repeat the experiment. The detailed materials and methods used to derive the data should remain in the section with that name.

Table 6. Characteristics of antibiotic-producing *Streptomyces*

Determination	*S. fluoricolor*	*S. griseus*	*S. coelicolor*	*S. nocolor*
Optimal growth temp (°C)	–10	24	28	92
Color of mycelium	Tan	Gray	Red	Purple
Antibiotic produced	Fluoricillinmycin	Streptomycin	Rholmondelay[a]	Nomycin
Yield of antibiotic (mg/ml)	4,108	78	2	0

[a]Pronounced "Rumley" by the British.

Table 7. Characteristics of antibiotic-producing *Streptomyces*

Organism	Optimal growth temp (°C)	Color of mycelium	Antibiotic produced	Yield of antibiotic (mg/ml)
S. fluoricolor	–10	Tan	Fluoricillinmycin	4,108
S. griseus	24	Gray	Streptomycin	78
S. coelicolor	28	Red	Rholmondelay[a]	2
S. nocolor	92	Purple	Nomycin	0

[a]Where the flying fishes play.

Table 8. Induction of creatinine deaminase in *Clostridium* sp. strains XP32 and XP56

	Clostridium sp. strain XP32		*Clostridium* sp. strain XP56	
N source[a]	Total enzyme[b]	Sp act (U/mg of protein)	Total enzyme	Sp act (U/mg of protein)
Ammonia	0.58	0.32	0.50	0.28
Glutamic acid	5.36	1.48	2.18	0.61
Aspartic acid	2.72	0.15	1.47	0.06
Arginine	3.58	2.18	3.38	2.19
Creatinine	97.30	58.40	104.00	58.30

(From the 2010 instructions to authors, Journal of Bacteriology, 2010, 192(1):1–20. Reprinted with permission from the American Society for Microbiology.)

[a]The inoculum was grown in glucose broth with ammonium sulfate, washed twice, and then transferred into the media with the N sources listed above.

[b]Enzyme units in cell extract obtained from ca.10^{10} cells.

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Note that these tables have three horizontal rules (lines) but no vertical rules. Virtually all tables are constructed this way. Occasionally, straddle rules (as below "sp. strain XP32" and "sp. strain XP56" in Table 8) are used. Vertical rules normally are not used in tables.

EXPONENTS IN TABLE HEADINGS

If possible, avoid using exponents in table headings. Confusion has resulted because some journals use positive exponents and some use negative exponents to mean the same thing. For example, some have used "cpm $\times$ 10^3" to refer to thousands of counts per minute, whereas others have used "cpm $\times$ 10^{-3}" for the same thousands of counts. If it is not possible to avoid such labels in table headings (or in figures), it may be worthwhile to state in a footnote (or in the figure legend), in words that eliminate the ambiguity, what convention is being used.

FOLLOWING THE JOURNAL'S INSTRUCTIONS

Instructions to authors commonly include a section about tables. Before preparing your tables, check the instructions to authors of your target journal. These instructions may indicate such items as the dimensions of the space

available, the symbols or form of lettering for indicating footnotes to tables, and the electronic tools to use in preparing tables. Looking at tables in the journal as examples also can aid in preparing suitable tables.

Traditionally, journals have asked authors to submit each table on a separate page at the end of the text. In addition, some journals have said to identify in the margin of the text the first mention of each table, for example, by writing "Table 3" and circling it. This procedure helps make sure that the author has indeed cited each table in the text, in numerical order. It also indicates to the compositor, at the page makeup stage, where to break the text to insert the tables. Today, some journals ask authors to embed tables in the text near their first mention. To determine whether tables should be placed within the text or at the end (and how, if requested, to indicate their placement), consult the instructions to authors.

TITLES, FOOTNOTES, AND ABBREVIATIONS

The title of the table (or the legend of a figure) is like the title of the paper itself. That is, the title or legend should be concise and not divided into two or more clauses or sentences. Unnecessary words should be omitted.

Give careful thought to the footnotes to your tables. If abbreviations must be defined, you often can give all or most of the definitions in the first table. Then later tables can carry the simple footnote: "Abbreviations as in Table 1."

Note that "temp" (Tables 1, 2, 6, and 7) is used as an abbreviation for "temperature." Because of space limitations in tables, almost all journals encourage abbreviation of certain words in tables that would not be abbreviated in the text. Capitalize any such abbreviations used as the first word in a column heading; do not use periods (except after "no.," which might be misread without the period). To identify abbreviations that your target journal considers acceptable in tables, you can look at tables published in the journal. Also, some journals list in their instructions to authors the abbreviations that can be used without definition in tables that they publish.

ADDITIONAL TIPS ON TABLES

The following are some further tips to help ensure that you design and use tables effectively.

Use wording that will be clear without reference to the text. For example, a table should not just refer to "Group 1" and "Group 2." Rather, it should identify each group by a more meaningful designation (examples: "High-Dose Group" and "Low-Dose Group," "REM Sleep Group" and "NREM Sleep Group," and "Graduate Students" and "Professors").

If a paper includes a series of tables presenting analogous data, use an analogous format for each. For example, if several tables compare the same four groups from different standpoints, list the four groups in the same order in each table. Or if different tables present data on the same variables at different times, keep listing the variables in the same order. Such consistency saves readers effort. (And it is easier for you, too.)

Finally, remember to mention every table in the text. And do so as soon as readers are likely to want to see the table. You have gone to the effort of preparing good tables. Be sure that readers can benefit fully from them.

CHAPTER 17

How to Prepare Effective Graphs

A good illustration can help the scientist to be heard when speaking, to be read when writing. It can help in the sharing of information with other scientists. It can help to convince granting agencies to fund the research. It can help in the teaching of students. It can help to inform the public of the value of the work.

—Mary Helen Briscoe

WHEN NOT TO USE GRAPHS

In the previous chapter, we discussed certain types of data that should not be tabulated. They should not be turned into graphs either. Basically, graphs are pictorial tables.

The point is this. Certain types of data, particularly the sparse type or the type that is monotonously repetitive, do not need to be brought together in either a table or a graph. The facts are still the same: Preparing and printing an illustration can be time-consuming and expensive, and you should consider illustrating your data only if the result is a real service to the reader.

This point bears repeating because many authors, especially those who are still beginners, think that a table, graph, or chart somehow adds importance to the data. Thus, in the search for credibility, there is a tendency to convert a few data elements into an impressive-looking graph or table. Don't do it. Your more experienced peers and most journal editors will not be fooled; they will soon deduce that (for example) three of the four curves in your graph are simply the standard conditions and that the meaning of the fourth curve could have been stated in just a few words. Attempts to dress up scientific data are usually doomed to failure.

If there is only one curve on a proposed graph, can you describe it in words? Possibly only one value is really significant, either a maximum or a minimum; the rest is window dressing. If you determined, for example, that the optimum pH value for a particular reaction was pH 8.1, it would probably be sufficient to state something like, "Maximum yield was obtained at pH 8.1." If you determined that maximum growth of an organism occurred at 37°C, a simple statement to that effect is better economics and better science than a graph showing the same thing.

If the choice is not graph versus text but graph versus table, your choice might relate to whether you want to impart to readers exact numerical values or simply a picture of the trend or shape of the data. Rarely, there might be a reason to present the same data in both a table and a graph, the first presenting the exact values and the second showing a trend not otherwise apparent. (This procedure seems to be rather common in physics.) Most editors would resist this obvious redundancy, however, unless the reason for it was compelling.

An example of an unneeded bar graph is shown in Fig. 5. This figure could be replaced by one sentence in the text: "Among the test group of 56 patients who were hospitalized for an average of 14 days, 6 acquired infections."

When is a graph justified? There are no clear rules, but let us examine some indications for their effective use.

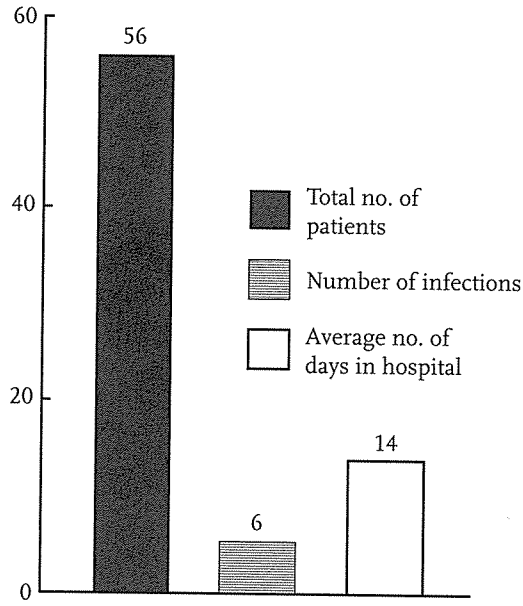


Figure 5. Incidence of hospital-acquired infections.

WHEN TO USE GRAPHS

Graphs are very similar to tables as a means of presenting data in an organized way. In fact, the results of many experiments can be presented either as tables or as graphs. How do we decide which is preferable? This is often a difficult decision. A good rule might be this: If the data show pronounced trends, making an interesting picture, use a graph. If the numbers just sit there, with no exciting trend in evidence, a table should be satisfactory (and perhaps easier and cheaper for you to prepare). Tables are also preferred for presenting *exact* numbers.

Examine Table 9 and Fig. 6, both of which record exactly the same data. Either format would be acceptable for publication, but Fig. 6 clearly seems superior to Table 9. In the figure, the synergistic action of the two-drug combination is immediately apparent. Thus, the reader can quickly grasp the significance of the data. It also appears from the graph that streptomycin is more effective than is isoniazid, although its action is somewhat slower; this aspect of the results is not readily apparent from the table.

HOW TO PREPARE GRAPHS

Early editions of this book included rather precise directions for using graph paper, India ink, lettering sets, and the like. Graphs had been prepared with these materials and by these techniques for generations.

Now we live in a world revolutionized by the computer. However, the *principles* of producing good graphs have not changed. The sizes of the letters and symbols, for example, must be chosen so that the final published graph in the journal is clear and readable.

The size of the lettering must be based on the anticipated reduction that will occur in the publishing process. This factor can be especially important if

Table 9. Effect of streptomycin, isoniazid, and streptomycin plus isoniazid on *Mycobacterium tuberculosis*[a]

	Percentage of negative cultures at:			
Treatment[b]	2 wk	4 wk	6 wk	8 wk
Streptomycin	5	10	15	20
Isoniazid	8	12	15	15
Streptomycin + isoniazid	30	60	80	100

[a]The patient population, now somewhat less so, was described in a preceding paper (61).
[b]Highest quality available from our supplier (Town Pharmacy, Podunk, IA).

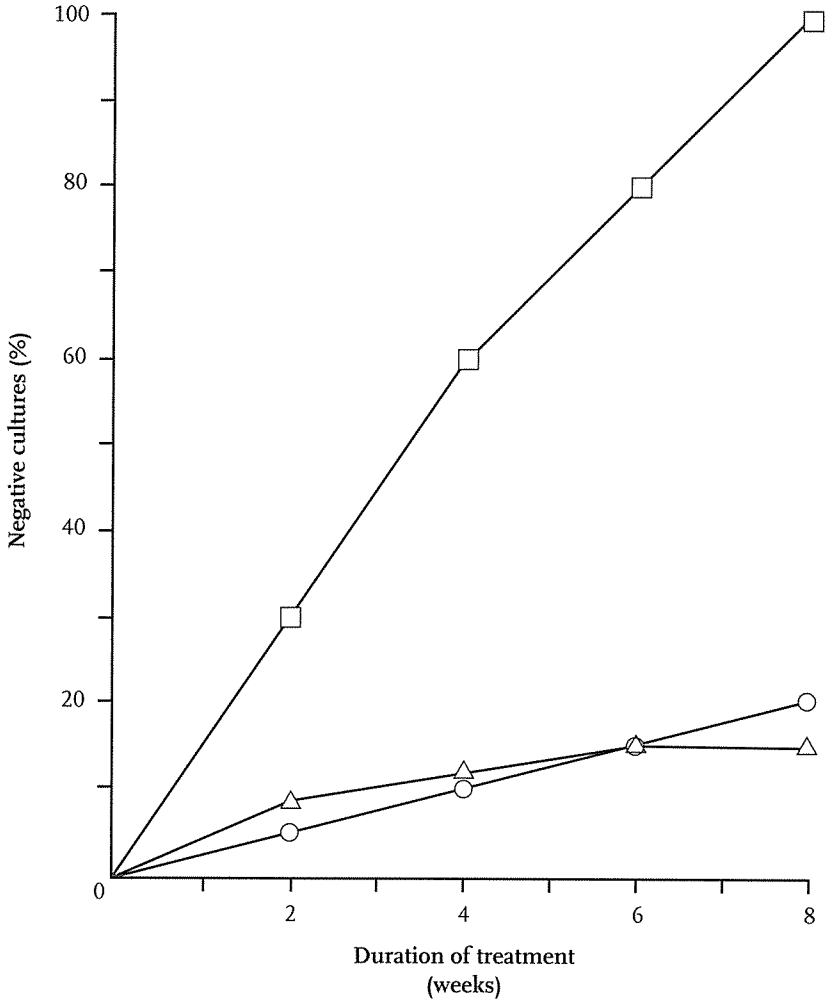


Figure 6. Effect of streptomycin (○), isoniazid (Δ), and streptomycin plus isoniazid (□) on *Mycobacterium tuberculosis.*

you are combining two or more graphs into a single illustration. Remember: Text that is easy to read on a large computer screen may become illegible when reduced to the width of a journal column.

Each graph should be as simple as possible. "The most common disaster in illustrating is to include too much information in one figure. Too much information in an illustration confuses and discourages the viewer" (Briscoe 1996).

Figure 7 is a nice graph. The lettering is large enough to read easily. It is boxed, rather than two-sided (compare with Fig. 6), making it a bit easier to estimate the values on the right-hand side of the graph. The scribe marks point inward rather than outward.

If your paper contains two or more graphs that are most meaningful when viewed together, consider grouping them in a single illustration. To maximize readability, place the graphs above and below each other rather than side by side. For example, in a two-column journal, placing three graphs in an "above and below" arrangement allows each graph to be one or two columns in width. If the graphs appear side by side, each can average only one third of a page wide.

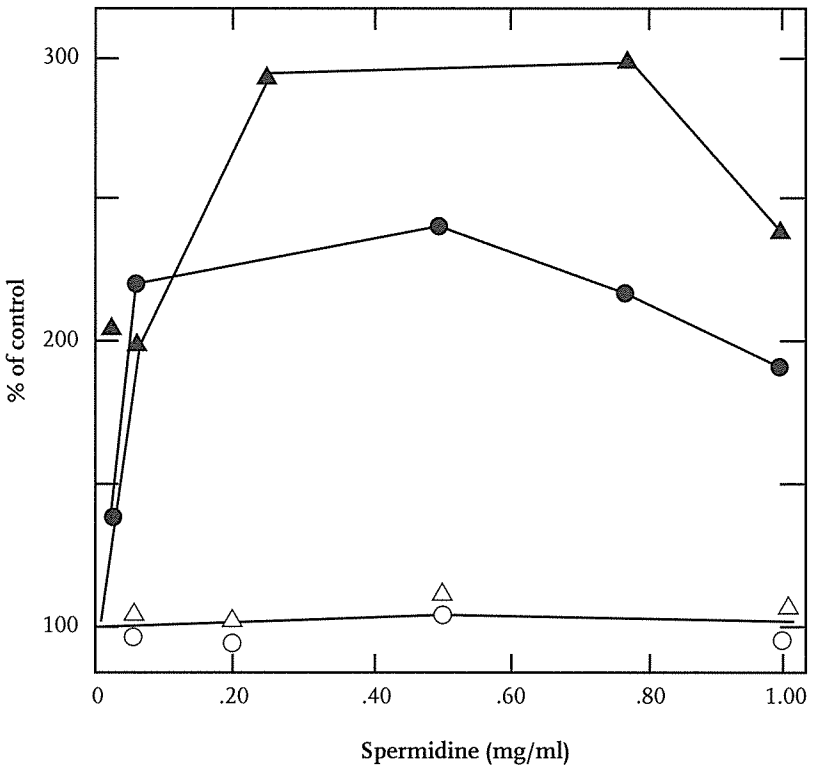


Figure 7. Effect of spermidine on the transformation of *B. subtilis* BR 151. Competent cells were incubated for 40 min with spermidine prior to the addition of 5 μg of donor DNA per ml (●) or 0.5 μg of donor DNA per ml (▲). DNA samples of 5 μg (○) or 0.5 μg per ml (△) were incubated for 20 min prior to the addition of cells. (Redrawn from Fig. 1 in Clark PO and Leach FR. Stimulation of *Bacillus subtilis* transformation by spermidine. Mol. Gen. Genet. 178:21–25, 1980. © by Springer-Verlag 1980. With kind permission of Springer Science and Business Media.)

Whether or not you group graphs in such a composite arrangement, be consistent from graph to graph. For example, if you are comparing interventions, keep using the same symbol for the same intervention. Also be consistent in other aspects of design. Both conceptually and aesthetically, the graphs in your paper should function as a set.

Do not extend the ordinate or the abscissa (or the explanatory wording) beyond what the graph demands. For example, if your data points range between 0 and 78, your topmost index number should be 80. You might feel a tendency to extend the graph to 100, a nice round number; this urge is especially difficult to resist if the data points are percentages, for which the natural range is 0 to 100. Resist this urge, however. If you do not, parts of your graph will be empty; worse, the live part of your graph will then be restricted in dimension, because you have wasted perhaps 20 percent or more of the width (or height) with empty white space.

In the preceding example (data points ranging from 0 to 78), your reference numbers should be 0, 20, 40, 60, and 80. You should use short index lines at each of these numbers and also at the intermediate 10s (10, 30, 50, 70). Obviously, a reference stub line halfway between 0 and 20 could only be 10. Thus, you need not letter the 10s, and you can then use larger lettering for the 20s, without squeezing. By using such techniques, you can make graphs simple and effective instead of cluttered and confusing.

SYMBOLS AND LEGENDS

If there is a space in the graph itself, use it to present the key to the symbols. In the bar graph (Fig. 5), the shadings of the bars would have been a bit difficult to define in the legend; given as a key, they need no further definition (and any additional typesetting, proofreading, and expense are avoided).

If you must define the symbols in the figure legend, you should use only those symbols that are considered standard and that are widely available. Perhaps the most standard symbols are open and closed circles, triangles, and squares (○, △, □, ●, ▲, ■). If you have just one curve, use open circles for the reference points; use open triangles for the second, open squares for the third, closed circles for the fourth, and so on. If you need more symbols, you probably have too many curves for one graph, and you should consider dividing it into two. Different types of connecting lines (solid, dashed) can also be used. But do *not* use different types of connecting lines *and* different symbols.

As to the legends, they should *normally* be provided on a separate page, not at the bottom or top of the illustrations themselves. The main reason is that the two portions commonly are processed separately during journal production. Consult the instructions to authors of your target journal regarding this matter and other requirements for graphs.

A FEW MORE TIPS ON GRAPHS

Design graphs, like tables, to be understandable without the text. For example, use meaningful designations (not just numbers) to identify groups. And refer to each graph as soon as readers are likely to want to see it. Do not leave readers trying to visualize your findings by sketching them on a napkin—only to find three pages later that a graph displays them.

Use graphs that depict your findings fairly and accurately. For example, do not adapt the scales on the axes to make your findings seem more striking than they are. With rare exceptions, avoid beginning a scale at anything other than zero. And if you interrupt a scale line to condense a graph, make the interruption obvious. Also, if the standard deviation is the appropriate way to show the variability in your data, do not substitute the standard error of the mean, which might make the data seem more consistent than it is.

Note that some journals (mainly the larger and wealthier ones) redraw graphs and some other types of figures to suit their own format. Whether or not a journal will do so, prepare your graphs well. Doing so will help make your findings and their value clear and will help show the care with which you do your work.